

Syed Sabbih Haider Shah

Senior AI Research Engineer

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Summary

Senior AI research engineer with 7+ years of experience building evaluated machine-learning systems for weather, geospatial analysis, and production applications. Work spans distributed model training, forecast verification, rare-event calibration, and production inference on NERSC Perlmutter, ORNL Frontier, and cloud platforms. Hired and managed a 15+ person cross-functional team and managed a separate 5+ person research and engineering team while remaining hands-on with Python, PyTorch, and deployment.

Core Skills

Machine learning: PyTorch, Transformers, U-Nets, diffusion models, LoRA/QLoRA, cross-encoders, spatiotemporal forecasting, computer vision, ASR/TTS, calibration and evaluation

Systems and data: Python, C++, CUDA, NumPy, pandas, xarray, Dask, Zarr, ONNX, NVIDIA DeepStream, Docker, Git, Linux, Slurm

Infrastructure: Distributed multi-GPU training and inference, NERSC Perlmutter, ORNL Frontier, Azure, GCP, streaming and batch pipelines

Professional Experience

Vayuh.ai

Senior AI Research Engineer — Full-time

Jan 2022–Present
United States / Remote

- Train and evaluate spatiotemporal weather models across sub-seasonal, seasonal, and decadal horizons, including billion-parameter research workloads on Perlmutter and Frontier.
- Built peril-specific U-Net workflows over a 1980–2025 atmospheric archive, with diagnostics for spatial agreement, hotspot capture, ranking, and probability calibration.
- Built and audited a 50,000-year, 3 km thunderstorm-wind catalog aligned with hail and tornado simulation years: 155.1 million parent storms, 28.36 billion event-cell rows, and 515 GB of Parquet output.
- Developed catastrophe-model calibration and loss-cost validation using NOAA event and gust records, correlated multi-peril sampling, and FEMA Hazus vulnerability assumptions.

Editable AI

AI Research Lead — Full-time

Feb 2025–Present
Remote

- Demonstrated Hyper-CONUS v2 achieved lower four-lead mean RMSE than HRRR for 2024 2 m temperature and dew point; for observed anomalies of at least +5 K and at most -5 K, temperature RMSE was 11.2% and 16.3% lower, respectively.
- Built a reproducible verification pipeline across 702 matched initialization times, eight forecasting systems, native and common grids, URMA analyses, and MRMS radar estimates.
- Built a deterministic severe-weather reporting pipeline that combines MRMS, SWDI, Local Storm Reports, Storm Events, ASOS observations, radar coverage, and explicit evidence scoring into reviewable case packets.

Dashverse

Lead Research Engineer — Full-time

Apr 2023–Feb 2026
United States / Remote

- Managed a team of 5+ and built the production AI inference stack for multimodal storytelling, including language, vision, and controllable diffusion models.
- Curated and cleaned LLM training data; fine-tuned screenplay, dialogue, and character-memory models with LoRA, QLoRA, and full-parameter training; evaluated and integrated them into production.
- Automated repetitive steps in short-form episode creation with smaller helper models and agents, reducing the time to produce a three-minute episode by about 40%.

WeVoz Jul 2021–Sep 2023
Lead Machine Learning Engineer — Contract Italy / Remote

- Trained and deployed a custom Italian speech-recognition model achieving 4.1% word error rate and a punctuation model achieving 92% accuracy.
- Optimized inference for low-compute environments and implemented scalable file-based and real-time streaming audio pipelines.
- Generated synthetic noisy speech data to improve robustness under deployment conditions.

Pagarba Solutions Sep 2021–Jan 2023
Team Lead, AI, MLOps & Optimization — Full-time United States / Remote

- Hired and managed a 15+ person team of researchers, MLOps engineers, mobile developers, and software developers; owned cross-functional ML projects from requirements through deployment.
- Built and deployed a 30,000-class LEGO recognition system with 96% top-3 accuracy and optimized models for constrained devices.

Additional Experience

STech.ai Sep 2019–Jan 2022
Machine Learning Engineer — Full-time Islamabad, Pakistan

- Led an RGB-only facial anti-spoofing system from research through deployment, achieving 97.6% accuracy without depth input.
- Built and optimized real-time NVIDIA DeepStream pipelines, custom plugins, and CUDA-based GPU components.

Teachinguide Jan 2021–Nov 2021
Data Scientist — Contract Switzerland / Remote

- Reduced model inference and deployment spend by 70% while implementing automated deployment, data logging, and metric monitoring.
- Deployed revenue forecasting at 4% mean absolute error; course classification at 96% precision and 95% recall; and a Transformer multi-topic classifier at 94% accuracy.

Selected Public Work

Forecast verification and storm evidence Weather / Geospatial ML

- Public methods and case studies: forecast verification and property-level storm evidence.

AI-agent prompt-injection detector AI security / NLP

- Built a context-aware cross-encoder and curated 13,421 adversarial and hard-negative pairs across agent and RAG attack surfaces; published dataset-specific evaluations without pooling model variants.

Severe-weather event-set generation Open source / Geospatial ML

- Implemented ensemble hail-magnitude sampling and multi-year Zarr processing workflows for reproducible stochastic event catalogs: github.com/sabbih-shah/eventset-generation.

Education

Master of Engineering, Machine Learning and Image Processing 2017–2019
Air University

Bachelor of Electrical Engineering (Electronics) 2013–2017
Air University